Cryptography

Lecture 2: Stream Ciphers: Theoretical Constructions

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Learning Objectives

By the end of this lecture, you will be able to:

1. Define semantic security and understand its relaxation from perfect security
2. Analyze semantic security games and compute adversarial advantages
3. Explain pseudorandom generators and their role in stream ciphers
4. Prove the security of stream ciphers using game-hopping techniques
5. Identify practical vulnerabilities of stream ciphers (two-time pad, malleability)
6. Understand methods for combining PRGs (parallel and sequential constructions)

Lecture Overview

Today we will cover:

1. Semantic Security - Relaxing perfect security
2. Pseudorandom Generators - Extending short keys
3. Stream Cipher Security - Theory and limitations

Semantic Security

From Perfect to Semantic Security

Recall: Perfect security requires $|\mathcal{K}| \geq |\mathcal{M}|$

- Impractical for real-world applications
- One-time pad needs keys as long as messages

Semantic Security: A practical relaxation

- Allows negligible information leakage about plaintext
- Ciphertext should not help distinguish between two chosen plaintexts
- Enables shorter keys through computational assumptions

Key insight: Perfect security = no information leaked

Semantic security = negligible information leaked

Semantic Security Game: Experiment Version

Two-experiment approach: Adversary $\mathcal{A}$ tries to distinguish encryptions

Experiment 0:

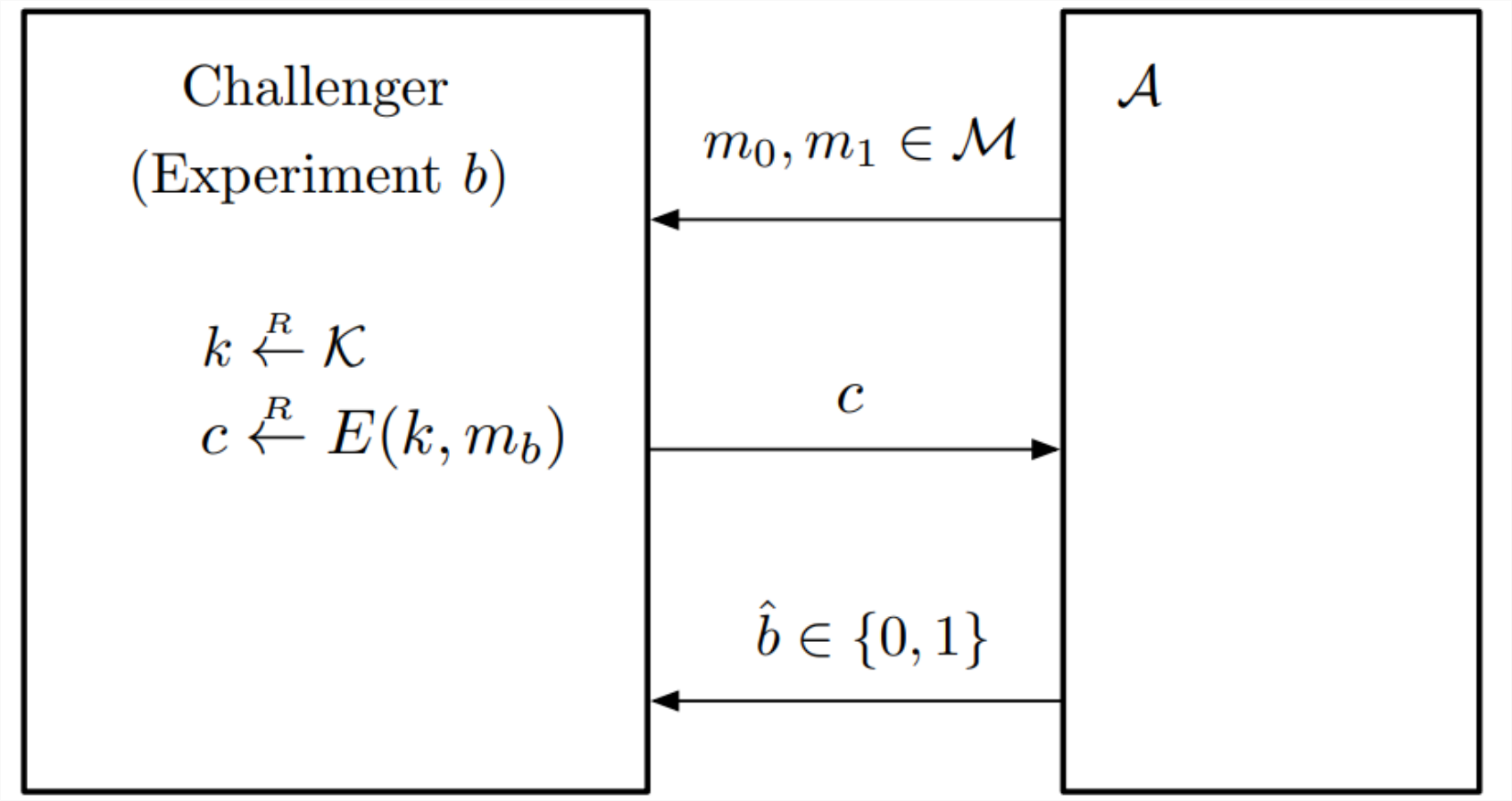
1. $m_0, m_1 \xleftarrow{\$} \mathcal{A}$
2. $k \xleftarrow{\$} \mathcal{K}$
3. $c \xleftarrow{\$} E(k, m_0)$
4. $\hat{b} \xleftarrow{\$} \mathcal{A}(c)$

Experiment 1

1. $m_0, m_1 \xleftarrow{\$} \mathcal{A}$
2. $k \xleftarrow{\$} \mathcal{K}$
3. $c \xleftarrow{\$} E(k, m_1)$
4. $\hat{b} \xleftarrow{\$} \mathcal{A}(c)$

Adversarial advantage: $\mathrm{Adv}_{\mathcal{E}}^{SS}(\mathcal{A}) \coloneqq |\Pr[W_0] - \Pr[W_1]|$

where W_b = event that adversary outputs 1 in experiment b


Challenger
(Experiment b)
$k \xleftarrow{R} \mathcal{K}$
$c \xleftarrow{R} E(k, m_b)$
$\mathcal{A}$
$m_0, m_1 \in \mathcal{M}$
c
$\hat{b} \in \{0, 1\}$

Semantic Security Game: Bit-Guessing Version

Equivalent single-game formulation:

Bit-Guessing Game:

1. $m_0, m_1 \xleftarrow{\$} \mathcal{A}$
2. $b \xleftarrow{\$} \{0,1\}$
3. $k \xleftarrow{\$} \mathcal{K}$
4. $c \xleftarrow{\$} E(k, m_b)$
5. $\hat{b} \xleftarrow{\$} \mathcal{A}(c)$
6. Return $(b = \hat{b})$

Adversarial advantage: $\mathrm{Adv}_{\mathcal{E}}^{SS}(\mathcal{A}) \coloneqq 2|\Pr[W] - \frac{1}{2}|$

where W = event adversary wins the bit-guessing game

Security Definition

Formal security requirement:

A cryptosystem is semantically secure if for all efficient adversaries $\mathcal{A}$:

$\mathrm{Adv}^{SS}(\mathcal{A})$ is negligible

Key definitions:

- Efficient adversary: Probabilistic polynomial-time (PPT) algorithm
- Negligible function: $f(n) = \mathcal{O}\left(\frac{1}{g(n)}\right)$ for super-polynomial $g(n)$

Intuition: No computationally bounded adversary can gain significant advantage in distinguishing encryptions of chosen plaintexts

Practical meaning: Security relies on computational hardness assumptions rather than information-theoretic impossibility

Pseudorandom Generators and Stream Ciphers

The Key Extension Problem

One-time pad limitations:

- Requires key length ≥ message length
- Key distribution and storage challenges
- Perfect security but impractical

Desired solution:

- Use short ℓ-bit seed s as key
- Extend to long L-bit keystream
- Maintain security properties

Stream cipher construction:

$$E(s, m) \coloneqq G(s) \oplus m$$

$$D(s, c) \coloneqq G(s) \oplus c$$

where $G : \{0,1\}^{\ell} \to \{0,1\}^{L}$ is a pseudorandom generator

Pseudorandom Generator Definition

Formal definition:

A pseudorandom generator (PRG) G is a deterministic algorithm:

- Input: seed s from finite seed space $\mathcal{S}$
- Output: r from finite output space $\mathcal{R}$
- Requirement: $|\mathcal{R}| > |\mathcal{S}|$ (expansion)

Typical case: $\mathcal{S} = \{0,1\}^{\ell}$, $\mathcal{R} = \{0,1\}^{L}$ where $L > \ell$

Key property: Output should be computationally indistinguishable from truly random strings

Security goal: No efficient adversary can distinguish $G(s)$ from random $r \xleftarrow{\$} \mathcal{R}$

PRG Security Game

Security formalized through distinguishing game:

PRG Security Game:

1. $s \xleftarrow{\$} \mathcal{S}$ (random seed)
2. $r_0 \leftarrow G(s)$ (PRG output)
3. $r_1 \xleftarrow{\$} \mathcal{R}$ (truly random)
4. $b \xleftarrow{\$} \{0,1\}$ (random choice)
5. $\hat{b} \xleftarrow{\$} \mathcal{A}(r_b)$ (adversary distinguishes)
6. Return $(b = \hat{b})$ (win if correct)

PRG advantage: $\mathrm{Adv}_G^{\mathrm{PRG}}(\mathcal{A}) \coloneqq 2|\Pr[W] - \frac{1}{2}|$

Security: G is secure if $\mathrm{Adv}_G^{\mathrm{PRG}}(\mathcal{A})$ is negligible for all efficient $\mathcal{A}$

Stream Cipher Security Analysis

Main Security Theorem

Theorem: If G is a secure PRG, then the stream cipher $\mathcal{E}$ constructed using G is semantically secure.

Quantitatively: For every adversary $\mathcal{A}$ against $\mathcal{E}$, there exists adversary $\mathcal{B}$ against G such that:

$$\mathrm{Adv}_{\mathcal{E}}^{\mathrm{SS}}(\mathcal{A}) \leq 2\mathrm{Adv}_{G}^{\mathrm{PRG}}(\mathcal{B})$$

Significance:

- Reduces stream cipher security to PRG security
- Provides concrete security bounds
- Justifies practical use of PRG-based encryption

Proof Technique: Game Hopping

Game hopping: Transform original security game through series of “hops” to simpler analysis

Game 0 - Original semantic security game:

1. $m_0, m_1 \xleftarrow{\$} \mathcal{A}$
2. $b \xleftarrow{\$} \{0,1\}$
3. $s \xleftarrow{\$} \mathcal{S}$
4. $r \leftarrow G(s)$
5. $c \leftarrow r \oplus m_b$
6. $\hat{b} \xleftarrow{\$} \mathcal{A}(c)$

Let W_0 = event that $\mathcal{A}$ wins Game 0

Then: $\mathrm{Adv}_{\mathcal{E}}^{SS}(\mathcal{A}) = 2|\Pr[W_0] - \frac{1}{2}|$

Game Hopping: The Transformation

Game 1 - Replace PRG with truly random:

1. $m_0, m_1 \xleftarrow{\$} \mathcal{A}$
2. $b \xleftarrow{\$} \{0,1\}$
3. $s \leftarrow \mathcal{S}$
4. $r \xleftarrow{\$} \mathcal{R}$
5. $c \leftarrow r \oplus m_b$
6. $\hat{b} \xleftarrow{\$} \mathcal{A}(c)$

Analysis:

- Game 1 = true one-time pad (perfect security)
- $\Pr[W_1] = \frac{1}{2}$ (adversary has no advantage)

Key insight: Difference between games relates to PRG security $|\Pr[W_0] - \Pr[W_1]| = \mathrm{Adv}_G^{\mathrm{PRG}}(\mathcal{B})$ for some adversary $\mathcal{B}$

Proof Completion

Constructing adversary $\mathcal{B}$ against PRG:

$\mathcal{B}$ receives input r (either $G(s)$ or random) and:

1. Runs $\mathcal{A}$ to get m_0, m_1
2. Chooses random bit b
3. Computes $c \leftarrow r \oplus m_b$
4. Gets $\hat{b} \leftarrow \mathcal{A}(c)$
5. Outputs 1 if $b = \hat{b}$, 0 otherwise

Analysis:

- If $r = G(s)$: $\mathcal{B}$ simulates Game 0
- If r random: $\mathcal{B}$ simulates Game 1
- Therefore: $|\Pr[W_0] - \Pr[W_1]| = \mathrm{Adv}_G^{\mathrm{PRG}}(\mathcal{B})$

Final bound: $\mathrm{Adv}_{\mathcal{E}}^{\mathrm{SS}}(\mathcal{A}) = 2|\Pr[W_0] - \frac{1}{2}|\, l = 2|\Pr[W_1] + \mathrm{Adv}_{G}^{\mathrm{PRG}}(\mathcal{B}) - \frac{1}{2}\,| = 2\mathrm{Adv}_{G}^{\mathrm{PRG}}(\mathcal{B})$

Stream Cipher Limitations

The Two-Time Pad Attack

Key reuse vulnerability: What happens with multiple messages?

Two messages with same key:

$$c_1 = m_1 \oplus G(s)$$

$$c_2 = m_2 \oplus G(s)$$

XOR the ciphertexts:

$$c_1 \oplus c_2 = (m_1 \oplus G(s)) \oplus (m_2 \oplus G(s)) = m_1 \oplus m_2$$

Consequence: Keystream cancels out, revealing relationship between plaintexts

Attack: Frequency analysis and pattern matching on $m_1 \oplus m_2$

Golden rule: Never reuse stream cipher keys!

Malleability Attack

Active adversary: Can modify ciphertexts in transit

Attack scenario:

- Original ciphertext: $c = m \oplus G(s)$
- Adversary wants to change message by known d
- Modified ciphertext: $c' = c \oplus d$

Result: $D(s, c') = c' \oplus G(s) = (c \oplus d) \oplus G(s) = m \oplus d$

Example: If adversary knows message contains “attack at dawn”, they can change it to “attack at dusk” by XORing with appropriate bit pattern

Limitation: Not a break of semantic security, but shows stream ciphers lack integrity protection

Security Assessment

Are stream ciphers still secure?

Semantic security perspective:

- Two-time pad: Not an attack on single-message security game
- Malleability: Not an attack on confidentiality
- Stream ciphers remain semantically secure under proper use

Practical implications:

- Stream ciphers provide confidentiality only
- Need additional mechanisms for integrity and authenticity
- Key management becomes critical

Usage guidelines:

- One key per message (or use nonces/IVs)
- Combine with authentication codes for integrity
- Implement proper key derivation and management

Extending Pseudorandom Generators

The Extension Problem

Scenario: Given secure PRG G, how to generate more pseudorandom bits?

Two main approaches:
1. Parallel construction - Run multiple PRGs simultaneously
2. Sequential construction - Chain PRGs together

Security question: How does security degrade with extension?

Key insight: Both constructions have linear security degradation - security decreases proportionally with number of PRG calls

Parallel Construction

Construction: Given PRG $G : \mathcal{S} \to \mathcal{R}$, build $G' : \mathcal{S}^n \to \mathcal{R}^n$

$G'(s_1, \ldots, s_n) \coloneqq (G(s_1), \ldots, G(s_n))$

Parallel PRG execution:

Seeds: $s_1, s_2, \ldots, s_n$ (independent)

Outputs: $G(s_1), G(s_2), \ldots, G(s_n)$

Final output: Concatenation of all outputs

Security bound: $\mathrm{Adv}^{\mathrm{PRG}}_{\{G'\}}(\mathcal{A}) \leq n \cdot \mathrm{Adv}^{\mathrm{PRG}}_{G}(\mathcal{B})$

Trade-off: $n\times$ more output, but $n\times$ security degradation

Sequential Construction

Construction: Chain PRGs by feeding output as next input

Given $G : \mathcal{S} \to \mathcal{S} \times \mathcal{R}$ (outputs seed + random bits)

Sequential PRG $G'(s)$:

$s_0 \leftarrow s$

for $i = 1$ to n:

$(r_i, s_i) \leftarrow G(s_{\{i-1\}})$

return $(r_1, \ldots, r_n, s_n)$

Security bound: Same as parallel case $\mathrm{Adv}^{\mathrm{PRG}}_{\{G'\}}(\mathcal{A}) \leq n \cdot \mathrm{Adv}^{\mathrm{PRG}}_{G}(\mathcal{B})$

Insight: Cost to stretch s bits to $s + n$ bits same as s to $s + 1$ bits (for polynomial n)

Proof Technique: Hybrid Argument

Hybrid argument: Prove security through sequence of games

For parallel construction with n PRGs:

- Hybrid 0: All n outputs from PRG
- Hybrid 1: First output random, remaining $(n-1)$ from PRG
- Hybrid 2: First two random, remaining $(n-2)$ from PRG
- …
- Hybrid n: All n outputs random

Key observation: Adjacent hybrids differ by single PRG vs random

Reduction: Adversary distinguishing hybrids $j-1$ and j gives PRG distinguisher

Final bound: $|p_n - p_0| \leq \sum_{\{j=1\}}^{n} |p_j - p_{\{j-1\}}| \leq n \cdot \mathrm{Adv}^{\mathrm{PRG}}(\mathcal{B})$

Lecture Summary

Key Takeaways: Theory

1. Semantic Security
- Practical relaxation of perfect security
- Allows negligible information leakage
- Enables shorter keys through computational assumptions

2. Pseudorandom Generators
- Deterministic expansion of short seeds to long keystreams
- Security based on computational indistinguishability
- Foundation for modern symmetric cryptography

3. Stream Cipher Security
- PRG security implies semantic security (with concrete bounds)
- Game-hopping: powerful proof technique for cryptographic reductions
- Theory provides rigorous foundation for practical constructions

Key Takeaways: Theory

4. Extension Methods
- Parallel and sequential constructions enable longer outputs
- Linear security degradation in number of PRG calls
- Hybrid arguments: standard technique for analyzing composed systems

5. Practical Limitations
- Key reuse completely breaks security (two-time pad)
- Malleability attacks possible without authentication
- Stream ciphers provide confidentiality only, not integrity

Central insight: Stream ciphers bridge theory and practice - rigorous security analysis enables confident real-world deployment

Looking Ahead: Next Lectures

Building Stream Ciphers in practice:

Questions?

Thank you

Stream Ciphers: Theoretical Constructions

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Next lecture: Stream Ciphers: Practical Constructions